Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH (ELECTRICAL & ELECTRONICS ENGINEERING)

MAKEUP EXAMINATIONS, DECEMBER 2022

Renewable Energy [ELE 4086]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 27 December 2022** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Determine the local apparent time corresponding to 12.30 IST on July 24 at Delhi ($28^0 35'$N, $77^0 30'$ E), and also find the declination angle, if the equation of correction is – (1'.06''). ***(03)***

1B. Explain the construction and working principle of a Pyranometer with a neat diagram. ***(04)***

1C. Explain the following: declination angle, Hour angle and latitude of location w.r.t Solar radiation geometry. ***(03)***

2A. At Nagpur, the following observations are made:

Theorical maximum possible sunshine hours = 9.5 h

Average measured length of day during April = 9.0 h

Solar radiation for clear day, H0= 2100 kJ/m^2/day

Constants: a=0.27, b=0.50

Calculate the average daily global radiation. ***(03)***

2B. Describe The parabolic concentrator collector with its neat diagram, list out its advantages. ***(03)***

2C. Describe the working principle of solar pond electric power plant with its neat diagram. ***(04)***

3A. Derive an equation for maximum power extraction from the wind mill. ***(03)***

3B. Describe the function of different parts of the horizontal axis wind mill with its neat diagram. List out the site selection criteria for wind mill installation. ***(04)***

3C. Explain the factors affecting the Generation of biogas in the biogas plant. ***(03)***

4A. Compare & discuss the working principle of fixed dome & floating drum in the biogas generation with neat diagrams. ***(04)***

4B. The following data are given for a family biogas digester suitable for the o/p of 8 cows; The retention time is 25 days , temp is 30^0 c , dry matter consumed per day is 2 kg, biogas yield is 0.24 m3 per kg. The efficiency of the burner is 65% , methane proportion is 0.7. Heat of combustion of methane is 28 MJ/$m^{3.}$ ***(03)***

4C. Describe the Environmental problems from the Geothermal power plants, list out its advantages, disadvantages & applications. ***(03)***

5A. Describe & compare the working principle of the open and closed cycle OTEC systems. List out the advantages & disadvantages of OTEC systems. ***(03)***

5B. Compare Single basin single effect. & single basin double effect tidal power plants with neat diagrams. ***(04)***

5C. The observed difference between the high and low water tide is 8.5 m for a proposed tidal site. The basin area is about 0.5 sq. km which can generate power for 3 hours in each cycle. The average available head is assumed to be 8 m, and the overall efficiency of the generation to be 70%. Calculate the power in H.P. at any instant and the yearly power output. Average specific weight of sea water is assumed to be 1025 kg/m^3. ***(03)***